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Oral care system

Abstract

The present disclosure generally relates to oral care systems and devices. An oral care system includes a container defining an interior portion and a moveable funnel configured to slide in the interior portion of the container. The moveable funnel includes a funnel portion having a base connected to a tube, where a distal end of the tube includes a cutting edge angled relative to a central axis of the tube. The system further includes a toothbrush supported by the tube, the toothbrush having a handle extending out from the container.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/070,201, filed Nov. 28, 2022, which is a continuation of U.S. patent application Ser. No. 17/713,392, filed Apr. 5, 2022, which is a continuation of U.S. patent application Ser. No. 17/008,245, filed Aug. 31, 2020, which is a continuation of U.S. patent application Ser. No. 15/813,051, filed Nov. 14, 2017, which claims priority to U.S. Provisional Patent Application No. 62/421,911, filed on Nov. 14, 2016, U.S. Provisional Patent Application No. 62/457,708, filed on Feb. 10, 2017, and U.S. Provisional Patent Application No. 62/519,049, filed on Jun. 13, 2017. All of the aforementioned applications are hereby incorporated by reference in their entireties.

BACKGROUND

(1) Standard toothbrush designs are not ideal for use in hospitals or other medical treatment facilities where patients may have compromised immune systems. In these settings, there is an increased risk of, for example, contracting pneumonia. Accordingly, compliance to a proper oral care regimen, which includes brushing three to four times per day, is important.

(2) Due to its repeated reuse, use of a standard toothbrush creates a risk of bacterial infection. Additionally, standard toothbrushes are also not suitable for some patients because patients may be unable to spit or otherwise voluntarily remove liquids from their mouths, may not have easy access to water, or may have difficulty holding and using a toothbrush having the standard size and shape. Standard toothpaste may also be unsuitable for use by such patients, and it may be difficult for such

patients to find and apply the proper amount of toothpaste to a toothbrush.

(3) In many circumstances, medical procedures like oral care must be repeated multiple times. For example, in oral care, a series of mouth care products must be used in a repetitious fashion, such as for periodic cleaning sessions, where there can be evacuation, brushing of the teeth, and swabbing of the mouth and gums. With the repetition of each procedure occurring after a predetermined interval, such as every few hours, it may be cumbersome for the medical care professional or the patient to gather the necessary instrument(s) repeatedly throughout the day.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following disclosure as a whole may be best understood by reference to the provided detailed description when read in conjunction with the accompanying drawings.

(2) FIG. 1 is a side perspective view of an oral care device, according to an exemplary embodiment.

(3) FIG. 2 is a sectional view of a unit dose oral fluid container, according to an exemplary embodiment.

(4) FIG. 3 is a sectional view of the unit dose oral fluid container of FIG. 2 with dimensions, according to an example embodiment.

(5) FIGS. 4A-4C are sectional views of the unit dose oral fluid container of FIG. 2 in various orientations, according to an exemplary embodiment.

(6) FIGS. 5A-5C are side views of a septum that may be included in the unit dose oral fluid container of FIG. 2, according to an exemplary embodiment.

(7) FIGS. 6A-6C depict steps of using an oral care system including the toothbrush of FIG. 1 and the unit dose oral fluid container of FIG. 2, according to an exemplary embodiment.

(8) FIG. 7 is a schematic side view of an oral care system, according to another exemplary embodiment.

(9) FIG. 8 is a schematic side view of an oral care system, according to another exemplary embodiment.

(10) FIG. 9 is a schematic side view of an oral care system, according to another exemplary embodiment.

(11) FIGS. 10A and 10B are schematic side views of an oral care system, according to another exemplary embodiment.

(12) FIG. 11A is a schematic side view of the unit dose oral fluid container of the oral care system of FIGS. 10A and 10B, according to an exemplary embodiment.

(13) FIG. 11B is a schematic top view of the unit dose oral fluid container of FIG. 11A, according to an exemplary embodiment.

(14) FIG. 12 is a schematic side view of an oral care system, according to another exemplary embodiment.

(15) FIG. 13 is a side view of a toothbrush, according to another exemplary embodiment.

(16) FIGS. 14A and 14B are schematic side views of an oral fluid bottle of FIG. 12, according to an exemplary embodiment.

(17) FIG. 15 is a side perspective view of a cylinder solution cup configured to fit into the oral fluid bottle FIGS. 14A and 14B, according to an exemplary embodiment.

(18) FIG. 16 is a side perspective view of a cover for the cylinder solution cup of FIG. 15, according to an exemplary embodiment.

(19) FIG. 17 is a schematic side view of an oral care system, according to another exemplary embodiment.

(20) FIGS. 18A-18C are side views of a toothbrush, according to another exemplary embodiment.

- (21) FIG. **18D** is a side perspective view of the toothbrush of FIGS. **18A-18C**, according to an exemplary embodiment.
- (22) FIG. **18E** is a top perspective view of the toothbrush of FIGS. **18A-18D**, according to an exemplary embodiment.
- (23) FIG. **19A** is a schematic side perspective view of an oral fluid bottle of FIG. **17**, according to an exemplary embodiment.
- (24) FIGS. **19B** and **19C** are schematic side views of the oral fluid bottle of FIG. **19A**, according to an exemplary embodiment.
- (25) FIG. **20A** is a schematic side perspective view of a cylinder solution cup configured to snap into the oral fluid bottle of FIGS. **19A-19C**, according to an exemplary embodiment.
- (26) FIGS. **20B** and **20C** are schematic side views of the cylinder solution cup of FIGS. **20A** and **20B**, according to an exemplary embodiment.
- (27) FIG. **20D** is a side perspective view of the cylinder solution cup of FIGS. **20A-20C** with a seal, according to an exemplary embodiment.
- (28) FIG. **21A** is a schematic side perspective view of a spittoon straw of FIG. **17**, according to an exemplary embodiment.
- (29) FIGS. **21B** and **21C** are side views of the spittoon straw of FIG. **21A**, according to an exemplary embodiment.
- (30) FIGS. **22A-22H** depict steps of using an oral care system including the toothbrush of FIGS. **18A-18E**, the oral fluid bottle of FIGS. **19A-19C**, the cylinder solution cup of FIGS. **20A-20C**, and the spittoon straw of FIGS. **21A-21C**, according to an exemplary embodiment.

DETAILED DESCRIPTION

- (31) According to various described embodiments, an oral care product that improves compliance with an oral care plan by providing all necessary tools and accessories for completing the plan without requiring significant effort from a patient or health care provider. According to various exemplary embodiments, an oral care system for use in a hospital setting includes an oral care device (e.g., a toothbrush as shown in FIG. **1**) and a unit dose oral fluid container (e.g., as shown in FIG. **2**). The oral care device and/or oral care system may be intended for single use. Various embodiments of an oral care systems are described in further detail below.
- (32) Referring first to FIG. **1**, a side perspective view of an oral care device, such as toothbrush **100** is shown, according to an exemplary embodiment. The toothbrush **100** includes a handle **102** coupled to a stem **104**, which in turn is coupled to a head **106**. The head **106** includes bristles **116** used to clean a patient's teeth. In some embodiments, the toothbrush **100** also includes supports **108** and **110** between the handle **102** and the stem **104** to provide length and stability to the toothbrush **100**. Further, in certain embodiments, the toothbrush **100** includes a suction mechanism. In such embodiments, the supports **108** and **110** house components of the suction mechanism. Suction is then provided by a device external to the toothbrush **100** and connected by tubing to the toothbrush **100**. Once the suction device external to the toothbrush **100** is activated, the patient may initiate suction at the head **106** of the toothbrush **100**. For example, the patient may cover a hole located on the handle **102** or press a button on the toothbrush **100** to provide suction.
- (33) Referring now to FIG. **2**, a sectional view of a unit dose oral fluid container **200** is shown, according to an exemplary embodiment. In certain embodiments, the oral fluid is an antiseptic fluid. The oral fluid container **200** defines a receptacle **202** in the interior of the oral fluid container **200**. The receptacle **202** is sized and configured to provide a stable base for the oral fluid container **200**. The receptacle **202** is also configured to serve as a repository and hold a volume of oral fluid **204**, such as an antiseptic fluid. In some embodiments, the receptacle **202** is configured to hold between 9.0 mL and 10.0 mL of oral fluid **204**, and in certain embodiments, the receptacle **202** is configured to hold 9.5 mL of oral fluid **204**. The receptacle **202** portion of the oral fluid container **200** may be cylindrical, though the receptacle **202** may have a different shape in various embodiments. A top portion of the oral fluid container **200** also includes a funnel **216** (e.g., a

projection having a mouth and a base, where the projection narrows from the mouth to the base, where the base is proximate the oral fluid **204**, and where the mouth comprises a shape that may be a circle, oval, square, rectangle, etc.), and an internal chamber **206** also within the interior of the oral fluid container **200** provides a channel between the receptacle **202** and the funnel **216**. In various arrangements, the receptacle **202** is coupled to the funnel **216** via a conical expansion region **226** (e.g., shaped like an inverted funnel).

(34) The internal chamber **206** has an hourglass shape and includes an inlet **208** and an outlet **210**. The inlet **208** provides a connection between the funnel **216** and the internal chamber **206**. The outlet **210** provides a connection between the internal chamber **206** and the receptacle **202**. The inlet **208** is defined by a first chamber **218** that is coupled to the funnel **216**. A second funnel portion **222** couples the first chamber **218** to a neck portion **214**, which has a reduced diameter relative to the first chamber **218** and a second chamber **220**. The outlet **210** is defined by the second chamber **220**. The second chamber **220** is coupled to the neck portion **214** via an inverted funnel portion **224**. Additionally, the outlet **210** includes a penetrable barrier that the toothbrush **100** must pass through for use of the oral fluid container. For example, in some embodiments, the outlet **210** is covered by a seal **212** prior to use. Alternatively, in other embodiments, the outlet **210** is covered by a septum (e.g., as described below with reference to FIGS. 5A-5C), a cover with a cross cut (e.g., similar to foam cover **666** or sponge cover **766** described below with reference to FIG. 9 and FIGS. 10A and 10B, respectively). The internal chamber **206**, and particularly second chamber **220**, provides a housing for the toothbrush **100** of FIG. 2A while not in use (e.g., as shown in FIG. 6A).

(35) The neck portion **214**, which has a reduced diameter relative to the first chamber **218** and second chamber **220**, is configured to remove excess oral fluid **204** from the toothbrush **100** as the toothbrush is removed from the internal chamber **206** for use (e.g., because, in the embodiment shown, the reduced diameter of the neck portion **214** presses out excess oral fluid **204** from the toothbrush **100** through a squeezing effect). The internal chamber **206** also facilitates passage of saliva and oral fluid waste that has been spit or expelled by the patient (hereinafter “brushing waste”), from the funnel **216** to the receptacle **202** during or after the use of the toothbrush **100** by the patient. The funnel **216** provides a larger target area for a patient to deposit the brushing waste while brushing to minimize the mess associated with spitting the used oral fluid into the receptacle **202**.

(36) In some embodiments, the oral fluid container **200** is manufactured using an injection mold. In other embodiments, rather than forming the neck portion **214** using injection molding, a separate element is placed inside the internal chamber **206** after injection molding to create a neck portion **214** having a reduced diameter. In still other embodiments, rather than forming the second funnel portion **222**, the neck portion **214**, and the inverted funnel portion **224** using injection molding along with the rest of the oral fluid container **200**, a separate element is placed inside the internal chamber **206** after injection molding of the rest of the oral fluid container **200** to create the second funnel portion **222**, the inverted funnel portion **224**, and the neck portion **214**. Additionally, in various embodiments, the oral fluid container **200** is configured to be symmetrical around an axis extending through a center of the oral fluid container **200**.

(37) In various embodiments, the unit dose oral fluid container **200** is designed to be spill-resistant. As such, in various arrangements, the unit dose oral fluid container **200** is designed with specific, spill-resistant dimensions. FIG. 3 is a sectional view of the unit dose oral fluid container of FIG. 2 with such dimensions, according to an exemplary embodiment. In the embodiment of FIG. 3, the oral fluid container **200** altogether has a total height of 11.4 cm (4.5 inches). From the base of the exemplary receptacle **202** to the base of the funnel **216**, the receptacle **202** is 8.9 cm (3.5 inches) tall, and the portion of the receptacle **202** with a constant diameter (i.e., not including the conical expansion region **226**) is 7.6 cm (3.0 inches) tall and has a diameter of 4.0 cm (1.5 inches). The funnel **216** has a height of 2.5 cm (1.0 inches) and an upper diameter of 5.1 cm (2.0 inches). The inlet **208** of the internal chamber **206**, and the first chamber **218** that defines the inlet **208**, has a

diameter of 1.9 cm (0.75 inches). The outlet **210**, and the second chamber **220** that defines the outlet, also has a diameter of 1.9 cm (0.75 inches), and the outlet **210** ends 2.5 cm (1.0 inches) above the base of the receptacle **202**. The first chamber **218** is 1.3 cm (0.5 inches) tall, and the second chamber **220** is 2.5 cm (1.0 inches) tall. The neck portion **214** has a height of 1.3 cm (0.5 inches) and a diameter of 1.3 cm (0.5 inches). The second funnel portion **222**, the neck portion **214**, and the inverted funnel portion **224** have a combined height of 2.5 cm (1.0 inches). Overall, the oral fluid container **200** has a wall width of 0.15 cm (0.06 inches). With such dimensions, the oral fluid container **200** is spill-resistant up to 15 mL to accommodate for a volume of both unused oral fluid and the saliva and oral fluid brushing waste (e.g., after use of the toothbrush **100**).

(38) Further referring to FIG. **3**, it should be understood that other dimensions may also be used to achieve the same or substantially the same ratios and configurations as those described above. Such dimension ratios allow the oral fluid container **200** to maintain its spill-resistant properties.

(39) Referring now to FIGS. **4A-4C**, schematic views of the unit dose oral fluid container **200** in various orientations that illustrate spill-resistant features of the oral fluid container **200** are shown, according to an exemplary embodiment. FIG. **4A** shows the oral fluid container **200** in an upright position with a combination **250** of unused oral fluid and brushing waste (e.g., after use of the toothbrush **100**), though the oral fluid container **200** could alternatively be filled with just oral fluid **204** (e.g., before use of the toothbrush **100**). FIG. **4B** shows the oral fluid container **200** in a horizontal position with the combination **250** of unused oral fluid and brushing waste. As shown, the combination **250** of unused oral fluid and brushing waste does not exit the oral fluid container **200** through the internal chamber **206** when the oral fluid container **200** is in the horizontal position but is instead maintained within the receptacle **202**. FIG. **4C** shows the oral fluid container **200** in an upside-down orientation. As shown, the combination **250** of unused oral fluid and brushing waste does not leave the oral fluid container **200** but is again maintained within the receptacle **202**.

(40) Referring now to FIGS. **5A-5C**, side views of a septum **300** are shown, according to an exemplary embodiment. The septum **300** includes a base **302** and a crown portion **304** forming an opening **306**. The crown portion **304** includes two or more flexible sections **308** separated by a slice in the septum material that defines the opening **306**. As shown in FIG. **5A**, in a natural state, the flexible sections meet together. The septum **300** as shown in FIG. **5A** is thus in a closed configuration. However, the flexible sections **308** are configured to separate and thereby open the opening **306** when subject to a force. For example, FIG. **5B** shows the toothbrush **100** entering through the opening **306** of the septum **300**. FIG. **5C** shows the head **106** of the toothbrush **100** having substantially passed through the opening **306** of the septum **300**.

(41) Accordingly, in some embodiments, the septum **300** is coupled to the internal chamber **206** of the oral fluid container **200** (e.g., at the inlet **208** or the outlet **210**). The flexible sections **308** allow the toothbrush **100** to enter and exit through the opening **306** of the septum **300**. However, because the septum **300** is naturally in a closed configuration, the septum **300** provides additional protection against spills, thereby increasing the spill-resistance of the oral fluid container **200**, while maintaining an opening for the toothbrush **100**. In some embodiments, the septum **300** also removes excess oral fluid **204** when the toothbrush **100** is removed from the internal chamber **206** by pulling the toothbrush **100** in the opposite direction through the septum **300** (e.g., because the two or more flexible sections **308** press excess oral fluid **204** from the toothbrush **100**).

(42) Referring now to FIGS. **6A-6C**, a method of using an oral care system **280** including a toothbrush (e.g., toothbrush **100**) and a unit dose oral fluid container (e.g., oral fluid container **200**) is depicted according to an exemplary embodiment. FIG. **6A** shows the oral care system **280** in a storage configuration with the toothbrush **100** held in the internal chamber **206** of the container **200**. As such, the seal **212** is still intact, and the receptacle **202** is filled with oral fluid **204**. While the embodiment shown in FIG. **6A** includes a seal **21** to prevent the toothbrush **100** from entering the oral fluid container **200** too early, the oral care system **280** may alternatively use any other mechanism that prevents the toothbrush **100** from being removed from the internal chamber **206**

and/or moved into the oral fluid **204** prior to use (e.g., such as the septum **300**).

(43) To begin using the oral care system **280**, the patient or care provider pushes the toothbrush **100** in the direction depicted by arrow **282** in FIG. **6B** to penetrate a penetrable barrier separating the toothbrush **100** from the oral fluid **204**. In the embodiment shown, the penetrable barrier is the seal **212**, though the oral fluid container **200** could additionally or alternatively include, for example, the septum **300**. The toothbrush **100** then becomes exposed to the oral fluid **204** in the receptacle **202** and can be pushed through the outlet **210** into the oral fluid **204** to become saturated. As shown in FIG. **4C**, the patient or care provider then pulls the toothbrush **100** in the direction depicted by arrow **284** in FIG. **6C** out of the internal chamber **206** to initiate oral treatment. As the toothbrush **100** is pulled out of the internal chamber **206**, the neck portion **214** squeezes the head **106** of the toothbrush **100** to remove excess oral fluid **204**. Additionally, in embodiments including a septum **300**, the flexible sections **308** of the crown portion **304** squeeze excess oral fluid **204** from the toothbrush **100**. The patient or care provider may then use the toothbrush **100** in a standard fashion to brush the patient's teeth. During or after brushing, the patient spits into the funnel **216** to remove oral fluid from the patient's mouth. When the patient has finished the oral care treatment, the oral care system **280** is to be disposed. In some arrangements, the patient or care provider reinserts the toothbrush **100** into the internal chamber **206** before disposing of the oral care system **280**.

(44) Referring now to FIG. **7**, a schematic side view of an oral care system **480** is shown, according to another exemplary embodiment. The oral care system **480** again includes a toothbrush, shown as the toothbrush **100** and a unit dose oral fluid container, shown as oral fluid container **400**. However, unlike the oral care system **280**, the oral care system **480** includes a separate repository container **403** and waste container **430**. As shown, the repository container **403** and the waste container **430** are connected via a bottom edge **432** and a top edge **434** and are separated by a gap **436** such that the repository container **403** and the waste container **430** do not interfere with one another. As shown in FIG. **7**, the repository container **403** and the waste container **430** are entirely separate containers, though in other embodiments, the repository container **403** and the waste container **430** may be at least partially connected (e.g., at the bottom of the repository container **403** and the waste container **430**).

(45) As shown, the repository container **403** defines a receptacle **402** in the interior of the repository container **403** configured to hold an oral fluid, such as antiseptic. For example, in some embodiments, the receptacle **402** is configured to hold 9.5 mL of oral fluid. The receptacle **402** is coupled to a top portion **438** with an inlet **408** via a conical extension **440**. The top portion **438** and the inlet **408** are sized to allow insertion and removal of the toothbrush **100**.

(46) The repository container **403** also includes an internal chamber **406**. Similar to the internal chamber **206**, the internal chamber **406** includes an outlet **410** providing a connection between the internal chamber **406** and the receptacle **402**. The internal chamber **406** is defined by a first chamber **418** that is coupled to the top portion **438**. A funnel portion **422** couples the first chamber **418** to a neck portion **414**. The outlet **410** is defined by an inverted funnel portion **424**, which is coupled to the neck portion **414** at the other end of the neck portion **414**. In some embodiments, the outlet **410** is covered by a seal **412** prior to use. In other embodiments, a seal or other type of penetrable barrier is located in another portion of the internal chamber **406** and/or the internal chamber **406** includes a septum (e.g., similar to the septum **300**). Similar to the internal chamber **206** of the oral fluid container **200**, the internal chamber **406** provides housing for the toothbrush **100** while not in use. Additionally, the neck portion **414** (and/or, in some embodiments, a septum) is configured to remove excess oral fluid **204** from the toothbrush **100** as the toothbrush is being removed from the internal chamber **406** for use (e.g., by pressing the excess oral fluid **204** from the head **106** of the toothbrush **100** through a squeezing effect).

(47) The waste container **430** includes a funnel **416** and a receptacle **442** that holds the brushing waste from a patient using the oral care system **480**. The funnel **416** is coupled to the receptacle **442** via a conical expansion region **426**. The funnel **416** provides a larger opening for a patient to

spit brushing waste into when the patient is using the oral care system **480**. The waste container **430** also includes an internal chamber **444**, coupled to the funnel **416** at a base **446** of the funnel **416**, that facilitates movement of the brushing waste from the funnel **416** to the receptacle **442**. The internal chamber **444** includes an outlet **448** through which the brushing waste passes into the receptacle **442**.

(48) In some embodiments, the waste container **430** and the repository container **403** are also dimensioned to be spill-resistant, similar to the oral fluid container **200** described above. In some embodiments, the dimensions and/or dimension ratios of the oral care system **480** are the same as the dimensions and/or dimension ratios of similar components in the oral fluid container **200** (e.g., as shown in FIG. 3).

(49) Referring now to FIG. 8, a schematic side view of an oral care system **580** is shown, according to another exemplary embodiment. The oral care system **580** includes the a toothbrush, shown as the toothbrush **100**, and a unit dose oral fluid container, shown as oral fluid container **500**. In various arrangements, the unit dose oral fluid container **500** of FIG. 8 is substantially similar to the unit dose oral fluid container **200** of FIG. 2. Accordingly, the oral fluid container **500** is shaped similarly to the oral fluid container **200** and includes a receptacle **502** and an internal chamber **506** in an interior portion of the oral fluid container **500** and a funnel **516** similar to the receptacle **202**, internal chamber **206**, and funnel **216** of the oral fluid container **200**. The receptacle **502** is coupled to the funnel **516** via a conical expansion region **526**. Further, similar to the internal chamber **206**, the internal chamber **506** has an hourglass shape with an inlet **508** defined by a first chamber **518** that is coupled to the funnel **516**, a second funnel portion **522** connecting the first chamber **518** to a neck portion **514**, and an inverted funnel portion **524** connecting the neck portion **514** to a second chamber **520** that defines an outlet **510**. As such, in various arrangements, the oral care system **580** is used similarly to the oral care system **280** described above.

(50) However, unlike the oral fluid container **200**, the unit dose oral fluid container **500** includes an oral fluid cup **560** configured to serve as a repository for the oral fluid **204**. As such, in the oral fluid container **200**, the oral fluid **204** is contained within the oral fluid cup **560** instead of residing in the receptacle **502**. In some embodiments, the oral fluid cup **560** is coupled to the unit dose oral fluid container **500**. For example, the oral fluid cup **560** is coupled to the container **500** by an attachment mechanism (e.g., glue) or an attachment device. Alternatively, the oral fluid cup **560** is constructed as a unitary piece with the container **500**. In other embodiments, the oral fluid cup **560** is sized to securely fit within the unit dose oral fluid container **502**, such as by a snap fit engagement.

(51) The oral fluid cup **560** includes a base **562** that holds the oral fluid **204**. In the embodiment shown in FIG. 8, the base **562** is cylindrical with a circular cross section. However, in other embodiments, the oral fluid cup **560** may have another cross-sectional shape, such as a circular cylinder or a square cylinder, or may have a varying diameter (e.g., be formed in a conical shape). A seal **512** extends across a top of the base **562**. In various arrangements, the seal **512** is made of a plastic material, a foil material, or a similar material that can be punctured with a small force. The seal **512** may span the entire diameter of the receptacle **502**, may span only the base **562**, or may span any portion therebetween. In the embodiment of FIG. 8, the seal **512** of unit dose oral fluid container **500** replaces the seal **212** that spans the outlet **210** of the unit dose oral fluid container **200**. As such, to use the oral care system **580**, the patient must press the toothbrush **100** against the seal **512** with enough force that it breaks or is otherwise breached. In some cases, manufacturing a unit dose oral fluid container with an oral fluid cup having a seal (e.g., as described herein with respect to FIG. 7, FIG. 8, and FIGS. 10A and 10B) provides manufacturing advantages over the unit dose oral fluid container **200** with the seal **212**. Alternatively, in other embodiments, the oral fluid cup **560** may include a different penetrable barrier, such as a cover with an internal cross cut.

(52) Referring now to FIG. 9, a schematic side view of an oral care system **680** is shown, according to another exemplary embodiment. The oral care system **680** includes a toothbrush, shown as

toothbrush **100**, and a unit dose oral fluid container, shown as oral fluid container **600**. In various arrangements, the unit dose oral fluid container **600** is configured somewhat similar or substantially similar to the unit dose oral fluid container **200** shown in FIG. 2. As such, the oral fluid container **600** includes a receptacle **602** in an interior of the oral fluid container and a funnel **616**. The oral fluid container **600** also includes a chamber **606**, though the chamber **606** is configured differently from the internal chamber **206** of the oral fluid container **200**. In particular, the chamber **606** of unit dose oral fluid container **600** differs from the previously described inner chambers in that it does not have a neck portion with a decreased diameter. Instead, the chamber **606** has a substantially cylindrical shape and a substantially uniform cross section extending between an inlet **608** and an outlet **610**. The inlet **608** provides a connection between the funnel **616** and the chamber **606**, and the outlet **610** provides a connection between the chamber **606** and the receptacle **602**. The uniform cross section of the chamber **606**, as well as the cross sections of the receptacle **602** and the funnel **616**, are in any shape that facilitates the use of the oral care system **680**, such as circular, elliptical, oblong, etc.

(53) Further, as shown in FIG. 9, the chamber **606** has a portion that is internal to the receptacle **602** and a portion that extends above the receptacle **602**. The portion of the chamber **606** that extends above the receptacle **602** is housed in a cylindrical neck portion **614**. The cylindrical neck portion **614** is coupled to and sized according to a base **646** of the funnel **616**, which also defines the inlet **608** of the chamber **606**. Additionally, the cylindrical neck portion **614** is coupled to and sized according to a top portion **638** of the receptacle **602**. In this way, the cylindrical neck portion **614** connects the funnel **616** to the receptacle **602**, thereby allowing liquid to flow from the funnel **616** through the cylindrical neck portion **614** and into the receptacle **602**. As such, the oral care system **680** may be used similarly to the oral care system **280** described above (e.g., by inserting the toothbrush **100**, through the chamber **606**, to be coated with oral fluid **204**, removing the toothbrush from the receptacle **602** for use, and spitting into the receptacle **602** via the funnel **616**).

(54) In some embodiments, the funnel **616**, the chamber **606**, and the receptacle **602** are manufactured as a single piece. In other embodiments, the funnel **616**, the chamber **606**, and/or the receptacle **602** are manufactured as two or more pieces. For example, the funnel **616** and the chamber **606** are manufactured as a single piece that slidably fits into the top portion **638** of the receptacle **602**. Additionally, in various arrangements, the receptacle **602** and the top portion **638** are manufactured as a single piece or as separate pieces. For example, the receptacle **602** is manufactured as an open-topped cylinder, and the top portion **638** is configured to snap into the open top of the receptacle **602**.

(55) Similar to unit dose oral fluid container **500**, the unit dose oral fluid container **600** also includes an oral fluid cup **660** configured to serve as a repository for the oral fluid **204**. As such, the oral fluid **204** is contained within the oral fluid cup **660** instead of residing in the receptacle **602**. In various embodiments, the oral fluid cup **660** is coupled to the unit dose oral fluid container **600** as described above with respect to the oral fluid cup **560** and the unit dose oral fluid container **500**. The oral fluid cup **660** includes a base **662** that holds the oral fluid **204**. In the embodiment of FIG. 9, the base **662** is cylindrical with a circular cross section. However, in other embodiments, the base **662** may take on various cross-sectional shapes, such as a circular cylinder or a square cylinder, or may have a varying diameter (e.g., be formed in a conical shape). Further, in the embodiment of FIG. 9, the oral fluid cup **660** is sized to hold between 1 and 15 mL of oral fluid **204**, such as oral fluid (e.g., sized to hold 7 mL of the oral fluid **204**).

(56) As shown in FIG. 9, a seal **612** also extends across a top of the base **662**. In various arrangements, the seal **612** is made of a plastic material, a foil material, or a similar material that can be punctured with a small force. Similar to the seal **512**, the seal **612** may span the entire diameter of the receptacle **602**, may span only the top of the base **662**, or may span any portion therebetween. In the embodiment of FIG. 9, the seal **612** of the unit dose oral fluid container **600** replaces the seal **212** that spans the outlet **210** of the unit dose oral fluid container **200**. As such, to

use the oral care system **680**, the patient must press the toothbrush **100** against the seal **612** with enough force that it breaks or is breached.

(57) As shown, the unit dose oral fluid cup **660** further includes a foam cover **666** positioned on top of the seal **612**. The foam cover **666** has an internal cross cut **668** extending through the foam cover **666**. In some embodiments, the foam cover **666** is a disk with a height of approximately 0.6 cm (0.25 inches). In other embodiments, the height of the foam cover **666** is in the range of approximately 0.25 to 2.5 cm (0.1 to 1.0 inches). The foam cover **666** is described herein as made of foam, but in other embodiments, the cover **666** is instead made of a different material (e.g., rubber, sponge, etc.). Without any force applied, the internal cross cut **668** of the foam cover **666** remains in a closed position. However, the internal cross cut **668** of the foam cover **666** is configured to separate and thereby open the foam cover **666** when subject to a substantially perpendicular force in either direction, for example, by a force provided by the toothbrush **100**. Accordingly, the foam cover **666** is coupled to the receptacle **602** and/or the top of the oral fluid cup **660** and allows the toothbrush **100** to enter and exit the oral fluid cup **660** through the foam cover **666**. In this way, the foam cover **666** serves as an additional penetrable barrier and provides additional protection against spills of the oral fluid **204** from the oral fluid cup **660**, thereby increasing the spill-resistance of the oral fluid container **600**, while maintaining an opening for the toothbrush **100**.

(58) In use, once the toothbrush **100** passes through the internal cross cut **668** of the foam cover **666**, the toothbrush is pressed against the seal **612** to break or breach the seal **612**, allowing the toothbrush **100** to access the oral fluid **204**. The toothbrush **100** is then removed from the oral fluid cup **660** via the internal cross cut **668** of the foam cover **666** for use. In some embodiments, the foam cover **666** also removes excess oral fluid **204** from the toothbrush head when the toothbrush **100** is removed from the cup **660** through foam cover **666** (e.g., by pressing out and/or absorbing the excess oral fluid **204** as the toothbrush **100** is drawn through the internal cross cut **668**). In some embodiments, the foam cover **666** further absorbs excess oral fluid **204** from the oral fluid cup **660**, for example, oral fluid **204** that comes into contact with the foam cover **666** because of tipping or sloshing of oral fluid along the walls of the oral fluid cup **660**.

(59) Referring now to FIGS. **10A** and **10B**, schematic side views of an oral care system **780** are shown, according to another exemplary embodiment. The oral care system **780** includes a toothbrush, shown as the toothbrush **100**, and a unit dose oral fluid container, shown as oral fluid container **700**. FIG. **11A** depicts a schematic side view of the unit dose oral fluid container **700**, and FIG. **11B** depicts a schematic top view of the unit dose oral fluid container **700**, according to an exemplary embodiment. As shown, in various arrangements, the unit dose oral fluid container **700** is configured similarly to the unit dose oral fluid container **600** of FIG. **9**. Accordingly, the oral fluid container **700** includes a receptacle **702** defined by the interior of the oral fluid container **700**, a funnel **716**, and a chamber **706** that does not have a neck portion with a decreased diameter. Instead, similar to the chamber **606**, the chamber **706** has a substantially cylindrical shape and a substantially uniform cross section extending between an inlet **708** and an outlet **710**. The inlet **708** provides a connection between the funnel **716** and the chamber **706**, and the outlet **710** provides a connection between the chamber **706** and the receptacle **702**. The uniform cross sections of the chamber **706**, as well as the cross sections of the funnel **716** and the receptacle **702**, are in any shape that facilitates the use of the oral care system **780**, such as circular, elliptical, oblong, etc.

(60) Further, similar to the chamber **606** of the oral fluid container **600**, the chamber **706** has a portion that is internal to the receptacle **702** and a portion that extends above the receptacle **702**. The portion of the chamber **706** that extends above the receptacle **702** is housed in a cylindrical neck portion **714**. The cylindrical neck portion **714** is coupled to and sized according to a base **746** of the funnel **716**, which also defines the inlet **708** of the chamber **706**. Additionally, the cylindrical neck portion **714** is coupled to and sized according to a top opening **738** defined in an expansion region of the receptacle **702**. The expansion region **726** is conical in shape (e.g., shaped like an

inverted funnel) and couples the portion of the receptacle **702** with a substantially constant diameter to the chamber **706**. In this way, the cylindrical neck portion **714** connects the funnel **716** to the receptacle **702**, thereby allowing liquid to flow from the funnel **716** through the cylindrical neck portion **714** and into the receptacle **702**. Thus, referring back to FIGS. **10A** and **10B**, the oral care system **780** is used similarly to the oral care system **280** described above (e.g., by inserting the toothbrush **100** through the chamber **706** to be coated with oral fluid **204**, removing the toothbrush from the receptacle **702** for use, and spitting into the receptacle **702** via the funnel **716**). Additionally, as with the unit dose oral fluid container **600**, the oral fluid container **700** is manufactured as a single piece or in multiple pieces.

(61) Referring back to FIGS. **10A** and **10B**, similar to the unit dose oral fluid containers **500** and **600**, the unit dose oral fluid container **700** also includes an oral fluid cup **760** configured to serve as a repository for the oral fluid **204**. As such, the oral fluid **204** is contained within the oral fluid cup **760** instead of residing in the receptacle **702**. In various arrangements, the oral fluid cup **760** is coupled to the unit dose oral fluid container **700** as described above with respect to the oral fluid cup **560** and the unit dose oral fluid container **500**. The oral fluid cup **760** includes a base **762** that holds the oral fluid **204**. In the embodiment of FIGS. **10A** and **10B**, the base **762** is cylindrical with a circular cross section. However, in other embodiments, the base **762** may take on various cross-sectional shapes, such as a circular cylinder or a square cylinder, or may have a varying diameter (e.g., be formed in a conical shape). Further, in various arrangements, the oral fluid cup **760** is sized to hold between 1 and 15 mL of the oral fluid **204**, such as oral fluid (e.g., sized to hold 7-7.5 mL of the oral fluid **204**).

(62) As shown in FIGS. **10A** and **10B**, a seal **712** also extends across a top of the base **762**, and, in various embodiments, the seal **712** is similar to the seals **512** and/or **612** described above. The unit dose oral fluid container **700** further includes a sponge cover **766** having an internal cross cut **768** through the foam cover. In various embodiments, the sponge cover **766** is configured and functions similarly to the foam cover **666** described above with respect to the unit dose oral fluid cup **660**, except that the sponge cover **766** is made of a sponge material rather than a foam material. In other arrangements, however, the sponge cover **766** is made of a non-sponge material (e.g., rubber, foam, etc.).

(63) Referring now to FIG. **12**, a schematic side view of an oral care system **880** is shown, according to another exemplary embodiment. The oral care system **880** includes a unit dose oral fluid container, shown as oral fluid container **800**, and a toothbrush, shown as toothbrush **900**. Referring to FIG. **13**, a side view of the toothbrush **900** is shown, according to an exemplary embodiment. The toothbrush **900** includes a handle **902** coupled to a lower stem **904**, which in turn is coupled to a head **906**. The handle **902** includes an upper stem **908** with disc-like projections **910** extending from the upper stem **908**. The diameters of the projections **910** vary along the length of the upper stem **908** to provide an ergonomically beneficial design. For example, the projections **910** provide a wider handle for the toothbrush so that patients with less gripping force or ability have an easier time gripping the toothbrush. As shown in FIG. **13**, in some embodiments, the projections **910** are wider near the bottom of the toothbrush **900** and narrower near the location where the lower stem **904** and upper stem **908** meet. As further shown in FIG. **13**, in certain embodiments, the projections **910** are also narrower near the center of the upper stem **908**.

(64) In various arrangements, the upper stem **908** is hollow to provide suction therethrough. Accordingly, as shown in FIG. **13**, the upper stem **908** includes a suction port **912** configured to couple to a suction mechanism. Further, the upper stem **908** includes a projecting suction opening **914**. The projecting suction opening **914** is positioned so that the patient can place a finger over the opening to activate suction through the upper stem **908**. The lower stem **904**, which extends between the upper stem **908** and the head **906**, is hollow to allow suction to travel between the head **906** and the suction port **912**.

(65) The head **906** includes bristles **916** and a foam section **918**. In some embodiments, the foam

section **918** is located on both the back and sides of the head **906**. In other embodiments, the foam section **918** is located on only one of the back or sides. Further, in some embodiments, the foam section **918** is glued to the head **906** of the toothbrush **900**, while in other embodiments the foam section **918** is attached to the head **906** through another mechanism (e.g., through an interference fit). As shown in FIG. **13**, in some embodiments, the foam section **918** has ridges to aid in cleansing. In various arrangements, the head **906** also includes at least one suction hole to provide for suction of fluids (e.g., saliva, antiseptic or other oral care fluid, etc.) from the mouth of the user. The at least one suction hole is located on a top, front, and/or back of the head **906**.

(66) In various arrangements, the toothbrush **900** is manufactured using cored out injection molding. In some embodiments, the toothbrush **900** is a unitary, single injection-molded piece forming the upper stem **908**, projections **910**, lower stem **904**, and the head **906**. In other embodiments, the toothbrush **900** is manufactured in separate pieces and assembled to form the completed toothbrush **900**. For example, the upper stem **908** and the lower stem **904** are formed together as a single injection-molded piece or formed as separate pieces that are later connected together.

(67) The toothbrush **900** is described herein with reference to the oral care system **880**. However, it should be understood that the toothbrush **900** may be used with any of the oral care systems described herein. Moreover, it should further be understood that the oral care system **880** may be used with a differently designed toothbrush, such as the toothbrush **100**.

(68) With respect to the oral care system **880**, as shown in FIG. **12**, the unit dose oral fluid container **800** includes an oral fluid bottle **801** and a cylinder solution cup **200**. Referring now to FIGS. **14A** and **14B**, schematic side views of the oral fluid bottle **801** are shown, according to an exemplary embodiment. The oral fluid bottle **801** includes a receptacle **802**, a funnel **816** provided on a top portion of the receptacle **802**, and an internal chamber **806** providing a channel between an interior of the receptacle **802** and the funnel **816**. As shown, the receptacle **802** and the internal chamber **806** are both contained within an interior portion of the oral fluid bottle **801**. The receptacle **802** is sized and configured to provide a stable base for the oral fluid bottle **801**. In the embodiment shown, the receptacle **802** of the oral fluid bottle **801** is cylindrical, though in other embodiments the receptacle **802** has another shape (e.g., a rectangular shape).

(69) The funnel **816** is coupled to the receptacle **802** at a base **846** of the funnel **816**. The funnel **816** facilitates the passage of brushing waste to an internal portion of the receptacle **802** during or after the use of a toothbrush (e.g., toothbrush **900**) by the patient. The funnel **816** provides a larger area for a patient to deposit the brushing waste into while brushing to minimize any mess associated with spitting into the oral fluid bottle **801**. As shown in FIGS. **14A** and **14B**, in some embodiments, the funnel **816** lies above the main body of the receptacle **802** and extends into the receptacle **802** to form the internal chamber **806**. The internal chamber **806** accordingly includes a second funnel portion **822** that narrows to a neck portion **814**. Liquid can be received into the internal chamber **806** through an inlet **808** at the top of the second funnel portion **822** and flow out of the internal chamber **806** via an outlet **810** at the bottom of the neck portion **814**. In certain embodiments, the receptacle **802**, funnel **816**, and internal chamber **806** are also a unitary body formed integral with one another. The oral fluid container **800** is configured to be used similarly to the oral fluid container **200** described above (e.g., by inserting the toothbrush **900** through the internal chamber **806** to be coated with oral fluid **204**, removing the toothbrush from the receptacle **802** for use, and spitting into the receptacle **802** via the funnel **816**). Accordingly, in various arrangements, the internal chamber **806** is sized to hold a toothbrush head (e.g., the toothbrush head **906**), and the neck portion **814** is sized to press out excess oral fluid **204** from the toothbrush head **906** when the toothbrush **900** is removed from the oral fluid bottle **801** by pulling the toothbrush **900** through the internal chamber **806**.

(70) Additionally, the receptacle **802** is configured to receive a cylinder solution cup through a lower portion of the receptacle **802**. As shown in FIGS. **14A** and **14B**, the receptacle **802** has an

open bottom for receiving the cylinder solution cup, though in other embodiments, the receptacle **802** is structured to receive the cylinder solution cup differently (e.g., include a slot for sliding in the cylinder solution cup). In various embodiments, the receptacle **802** also includes legs **872** defining the area that receives the cylinder solution cup. The legs **872** are spaced apart to create openings **874** for the ease of inserting the cylinder solution cup into the receptacle **802**. The receptacle **802** further includes apertures **870** for locking the cylinder solution cup into the receptacle **802**. The apertures **870** may have various arrangements on the receptacle **802**. For example, in some embodiments and as shown in FIGS. **14A** and **14B**, the receptacle **802** includes two sets of two apertures **870**, opposite one another, near the bottom portion of the receptacle **802**. (71) Referring now to FIG. **15**, a side perspective view of a cylinder solution cup **1000** is shown, according to an exemplary embodiment. The cylinder solution cup **1000** includes a cup base **1002** configured to serve as a repository for the oral fluid and receives brushing waste from a patient brushing his or her teeth. The cylinder solution cup **1000** is configured to hold a sufficient amount of oral fluid such that when the toothbrush **900** is dipped into oral fluid, the volume change in oral fluid due to displacement allows the head **906** of the toothbrush **900** to be substantially submerged in oral fluid while at the same time not spilling any of oral fluid out the top of the cup **1000**. In some embodiments, the cylinder solution cup **1000** is configured to hold approximately 7.5 mL of oral fluid.

(72) The cylinder solution cup **1000** includes a rim **1004** that defines a ledge **1006** and carries cup projections **1008**. The cup projections **1008** are spaced to align with the apertures **870** of the oral fluid bottle **801** such that the cup projections **1008** can snap into the apertures **870** to lock the cylinder solution cup **1000** into the receptacle **802** of the oral fluid bottle **801**. The cylinder solution cup **1000** also includes an upper portion **1010** that is wider than the cup base **1002**. The upper portion **1010** includes a gasket or o-ring **1014** that rests on the ledge **1006** to provide a seal between the cylinder solution cup **1000** and the receptacle **802** once the cylinder solution cup **1000** is snapped into the oral fluid bottle **801**.

(73) The upper portion **1010** defines an opening **1013** at a top of the cylinder solution cup **1000**. The opening **1013** holds a cover **1012**, which is illustrated with greater detail in FIG. **16**. The cover **1012** may be constructed of foam, rubber, or a similar flexible material and includes an internal cross cut **1016**. Similar to the internal cross cut **668** of the cover **666**, the internal cross cut **1016** allows the toothbrush **900** to enter and exit the cover **1012** and removes excess oral fluid from the toothbrush head **906**. The cover **1012** is glued or otherwise affixed to, or rested upon, the cylinder solution cup **1000**. In various embodiments, under the cover **1012**, the cylinder solution cup **1000** includes a foil seal (not shown) to prevent oral fluid from escaping the cylinder solution cup **1000** before use. The level of oral fluid contained in the cup base **1002** is accordingly selected to allow for ease of sealing the cylinder solution cup **1000** without spilling oral fluid. The seal is broken by pushing the head **906** of the toothbrush **900** through the internal cross cut **1016** of the cover **1012** and through the seal, if provided, after which the head **906** of the toothbrush **900** can be at least partially submerged in the oral fluid contained in the cup base **1002**. Accordingly, the seal and the cover **1012** serve as penetrable barriers that the toothbrush **900** must pass through in order to access the oral fluid.

(74) Referring back to FIG. **12**, the assembled oral care system **880** is shown. Accordingly, as can be seen through the opening **874** between the legs **872**, the cylinder solution cup **1000** has been inserted into the oral fluid bottle **801** through the open bottom of the oral fluid bottle **801**. The cylinder solution cup **1000** has also been snapped into the unit dose oral fluid container **800** via the apertures **870** and the cup projections **1008**. The head **906** of the toothbrush **900** sits in the neck portion **814** of the internal chamber **806**. The neck portion **814** of the internal chamber **806** is sized to hold the head **906** of the toothbrush **900** firmly in place, but the user may use a small force to move the head **906** of the toothbrush **900** within the internal chamber **806** for use. For example, to use the oral care system **880**, the patient presses the head **906** of the toothbrush **900** through the

outlet **810** of the internal chamber **806** into the receptacle **802**. The patient further presses the toothbrush **900** into the cylinder solution cup **1000** by providing a sufficient force to press through the cover **1012** and the seal, if provided, of the cylinder solution cup **1000**. The head **906** of the toothbrush **900** is then substantially covered by oral fluid contained within the cup base **1002** of the cup **1000** and removed from the receptacle **802** via the inlet **808** of the internal chamber **806**. As shown in FIG. **12**, the rim **1004** fits snugly against the walls of the receptacle **802**. Accordingly, the patient can spit into the receptacle **802** via the funnel **816**, and the tight fit between the cylinder solution cup **1000** and the receptacle **802** contains the brushing waste therein.

(75) In some embodiments, the oral fluid bottle **801** and the cylinder solution cup **1000** are sized and shaped to provide a spill-resistant design, similar to the unit dose oral fluid container **200** described above. In particular, in various arrangements, the receptacle **802**, funnel **816**, and internal chamber **806** are sized and configured, based upon the volume of the oral fluid in the solution cup and/or expected volume of brushing waste to be received, such that even with tipping of the receptacle **802** onto its side, no fluid or little fluid flows into the internal chamber **806** via the outlet **810** to be spilled outside of the oral fluid bottle **801**.

(76) Referring now to FIG. **17**, a schematic perspective side view of an oral care system **1180** is shown, according to another exemplary embodiment. The oral care system **1180** includes a unit dose oral fluid container, shown as oral fluid container **1100**, and a toothbrush, shown as toothbrush **1200**. Additionally, as shown, the unit dose oral fluid container **1100** includes an oral fluid bottle **1101**, a cylinder solution cup **1300**, and a spittoon straw **1400**. Referring to FIGS. **18A-18C**, side views of the toothbrush **1200** are shown, according to an exemplary embodiment. As shown, the toothbrush **1200** is configured similarly to toothbrush **900** shown in FIG. **13**. The toothbrush **1200** includes a handle **1202** coupled to a lower stem **1204**, which in turn is coupled to a head **1206**. The head **1206** includes bristles **1216** and a foam section **1218**. The handle **1202** includes an upper stem **1208** with disc-like projections **1210** extending from the upper stem **1208**.

(77) Like the projections **910**, the diameters of the projections **1210** vary along the length of the upper stem **1208** to provide an ergonomically beneficial design. However, unlike the projections **910**, the centers of the projections **1210** are offset from the upper stem **1208** such that the projections **1210** extend past the upper stem **1208** of the handle **1202** more in the direction of the bristles **1216** than in the direction of the foam section **1218** of the head **1206**, as shown in FIG. **18B**. Because the projections **1210** are offset from the upper stem, the toothbrush **1200** can be inserted into the spittoon straw **1400** as shown in FIG. **17** without crushing the bristles **1216**. Additionally, the projections **1210** include a projection seal **1220a** and a projection seal **1220b**. The projection seals **1220a** and **1220b** are configured to block a center conduit of the spittoon straw **1400** to prevent contaminants (e.g., dust) from entering the interior of the oral fluid container **1100**, as described in further detail below.

(78) In various arrangements, the upper stem **1208** is hollow to provide suction therethrough. Accordingly, the upper stem **1208** includes a suction port **1212** configured to couple to a suction mechanism, as shown in more detail in FIG. **18D** illustrating a side perspective view of the toothbrush **1200**. Further, the upper stem **1208** includes a projecting suction opening **1214**. The projecting suction opening **1214** is positioned and extends beyond the projections **1220** so that the patient can place a finger over the opening to activate suction through the upper stem **1208**. The lower stem **1204**, which extends between the upper stem **1208** and the head **1206**, is hollow to allow suction to travel between the head **1206** and the suction port **1212**.

(79) As noted above, the head **1206** includes bristles **1216** and a foam section **1218**. In some embodiments, the foam section **1218** is located on both the back and sides of the head **1206**, as shown in FIG. **18E** illustrating a top perspective view of the toothbrush **1200**. In other embodiments, the foam section **1218** is located on only one of the back or sides of the toothbrush head **1206**. Further, in some embodiments, the foam section **1218** is glued to the head **1206** of the toothbrush **1200**, while in other embodiments the foam section **1218** is attached to the head **1206**

through another mechanism (e.g., through an interference fit). As shown, in some embodiments, the foam section **1218** has ridges to aid in cleansing. In various embodiments, the head **1206** also includes one or more suction holes to provide for suction of fluids (e.g., saliva, antiseptic or other oral fluid, etc.) from the mouth of the user. The suction hole(s) are located on a top, front, and/or back of the head **1206**. In the embodiment shown in FIGS. **18A-18E**, the toothbrush **1200** includes a suction hole **1222a** formed transversely through the center of the head **1206** and a suction hole **1222b** formed on a top end of the toothbrush.

(80) Similar to the toothbrush **900**, in some embodiments, the toothbrush **1200** is manufactured using cored out injection molding. In other embodiments, the toothbrush is manufactured in separate pieces and assembled to form the completed toothbrush **1200**. Additionally, the toothbrush **1200** is described herein with reference to the oral care system **1180**. However, it should be understood that the toothbrush **1200** may be used with any of the oral care systems described herein. Moreover, it should further be understood that the oral care system **1180** may be used with a differently designed toothbrush, such as the toothbrush **100** or the toothbrush **900**.

(81) Referring now to FIGS. **19A-19C**, schematic side views of the oral fluid bottle **1101** are shown, according to an exemplary embodiment. The oral fluid bottle **1101** includes a receptacle **1102** and an internal chamber **1106** defined within an interior portion of the oral fluid bottle **1101**, where the internal chamber includes outlet **1110**. As shown in FIGS. **19A-19C**, the internal chamber **1106** includes a funnel portion **1122** that narrows to a neck portion **1114**. The internal chamber **1106** thus provides a channel for receiving the spittoon straw **1400**, as discussed in further detail below. Additionally, the funnel portion **1122** provides additional spill resistance to the oral fluid container **1100**. In the embodiment shown, the receptacle **1102** of the oral fluid bottle **1101** is configured as a hollow, circular cylinder, though in other embodiments the receptacle **1102** has another shape (e.g., a rectangular shape). Additionally, the oral fluid bottle **1101** is configured to be relatively tall compared to the other components of the oral care system **1180** and to widen from a top of the oral fluid bottle **1101** to the bottom, which provides stability for the oral fluid bottle **1101**.

(82) The receptacle **1102** is configured to receive the cylinder solution cup **1300** through a lower portion of the receptacle **1102**. As shown in FIG. **19A**, the receptacle **1102** has an open bottom portion for receiving the cylinder solution cup **1300**, though in other embodiments the receptacle **1102** is structured to receive the cylinder solution cup **1300** differently (e.g., include a slot for sliding in the cylinder solution cup **1300**). In various embodiments, the receptacle **1102** includes apertures **1170** for locking the cylinder solution cup **1400** into the receptacle **1102**. The apertures **1170** may have various arrangements on the receptacle **1102**. For example, in some embodiments, the receptacle **1102** includes two sets of two apertures **1170**, opposite one another, near the bottom portion of the receptacle **1102**.

(83) FIG. **20A** shows a schematic side perspective view of the cylinder solution cup **1300**, and FIGS. **20B** and **20B** show schematic side views of the cylinder solution cup **1300**, according to an exemplary embodiment. The cylinder solution cup **1300** includes a base **1302** configured to serve as a repository for an oral fluid and which receives brushing waste from a patient brushing his or her teeth. The base **1302** is configured to be wide enough to receive the end of the spittoon straw **1400**, as well as deep enough that the head **1206** of the toothbrush **1200** provided in the spittoon straw **1400** can be at least partially submerged in the oral fluid contained therein, as described in further detail below. Moreover, the cylinder solution cup **1300** is configured to hold a sufficient amount of oral fluid such that when the toothbrush **1200** and spittoon straw **1400** are dipped into the oral fluid, the volume change in the oral fluid due to displacement allows the head **1206** of the toothbrush **1200** to be substantially submerged in the oral fluid while at the same time not spilling any of the oral fluid out of the top of the cup **1300**. In some embodiments, the cylinder solution cup **1300** is configured to hold approximately 7.5 mL of oral fluid.

(84) The cylinder solution cup **1300** includes a rim **1304** that defines a ledge **1306** and carries

projections **1308**. The projections **1308** are spaced to align with the apertures **1170** of the oral fluid bottle **1101** such that the projections **1308** can snap into the apertures **1170** to lock the cylinder solution cup **1300** into the receptacle **1102** of the oral fluid bottle **1101**. Additionally, the rim **1304**, and the cylinder solution cup **1300** overall, are configured such that when the projections **1308** are snapped into the apertures **1170**, the receptacle **1102** of the oral fluid bottle **1101** and the rim **1304** of the cylinder solution cup **1300** can contain a liquid.

(85) Furthermore, the cylinder solution cup includes legs **1318** defining an internal space into which the base **1302** projects. The legs **1318** are spaced apart so as to provide the cylinder solution cup **1300** and, once the cylinder solution cup **1300** is snapped into the oral fluid bottle **1101**, the oral fluid container **1100** as a whole a stable base. The cylinder solution cup **1300** may include any number of legs **1318** that provide a stable base, though in the embodiment shown in FIGS. **20A-20C** the cylinder solution cup **1300** has four legs **1318**.

(86) The rim **1304** also defines an opening **1313** at a top of the cylinder solution cup **1300** that feeds into the base **1302**. In various embodiments, the opening **1313** is covered by a seal **1314**, as illustrated in FIG. **20D**. The seal **1314** is formed of a foil, plastic, or other frangible material that can be pierced or breached. The seal **1314** is configured to contain the oral fluid inside the base **1302** until the oral care system **1180** is to be used. However, the seal **1314** is also configured to be breached, such as by being pierced by the spittoon straw **1400** when a patient uses the oral care system **1180**, as described in further detail below. After the seal **1314** has been breached or pierced, the head **1206** of the toothbrush **1200** is at least partially submerged in the oral fluid contained within the base **1302**. Alternatively, in other embodiments, the opening **1313** is instead sealed through another penetrable barrier, such as a cover with a cross cut similar to the cover **1012** discussed above with respect to the cylinder solution cup **1000**.

(87) Referring now to FIG. **21A**, a schematic side perspective view of the spittoon straw **1400** is shown, according to an exemplary embodiment. The spittoon straw **1400** includes a funnel **1416** that connects at a base **1446** of the funnel **1416** to a tube **1490** with an opening **1492** at the end thereof. The funnel **1416** and the tube **1490** are configured to be inserted into the internal chamber **1106** of the oral fluid bottle **1101** such that the tube **1490** protrudes into the receptacle **1102**. Accordingly, the spittoon straw **1400** facilitates the passage of brushing waste into an internal portion of the receptacle **1102** during or after the use of a toothbrush (e.g., toothbrush **1200**) by the patient. The funnel **1416** provides a larger area for the patient to deposit the brushing waste into while brushing to minimize any mess associated with spitting into the unit dose oral fluid container **1100**. The funnel **1416** and the tube **1490** then channel the brushing waste into the receptacle **1102** via an inlet **1408** of the spittoon straw **1400** and the opening **1492**. Additionally, the end of the spittoon straw **1400** is cut at a diagonal, such that the plane of the opening **1492** lies at an angle relative to the axis of the straw **1400**. In this way, the end of the straw at the opening **1492** comprises a leading edge that forms a cutting portion **1494** on one side. The cutting portion **1494** can be used to break or breach the seal **1314** on the cylinder solution cup **1300** snapped into the oral fluid bottle **1101**.

(88) Referring now to FIGS. **21B** and **21C**, side views of the spittoon straw **1400** are shown, according to an exemplary embodiment. As shown in FIGS. **21B** and **21C**, the funnel base **1446** includes a lip such that the base **1446** is wider than the top of the tube **1490** of the spittoon straw **1400**. Accordingly, the spittoon straw **1400** is configured to receive the toothbrush **1200** in a central channel of the spittoon straw **1400** formed in the funnel **1416** and the tube **1490** such that the projection seal **1220a** rests on the base **1446**, thereby sealing the oral care system **1180** from outside contaminants once assembled. Additionally, the tube **1490** is sized with a diameter wide enough to snugly fit the projection seal **1220b** therein, further sealing the oral care system **1180** from outside contaminants. Furthermore, as shown, the tube **1490** includes an indentation **1495**. The indentation **1495** is configured to rest below the head **1206** of the toothbrush **1200** such that once the toothbrush **1200** is inserted into the tube **1490**, a small force must be applied to remove

the toothbrush **1200** from the spittoon straw **1400**.

(89) As shown in FIGS. **21A-21C**, the tube **1490** of the spittoon straw **1400** also includes projections **1496**. The projections **1496** are angled such that the spittoon straw **1400** can be inserted into the internal chamber **1106** of the oral fluid bottle **1101** but, once inserted, configured to abut against the outlet **1110** of the oral fluid bottle **1101** to prevent spittoon straw **1400** from being pulled out from the internal chamber **1106**. As such, the projections **1496** prevent the spittoon straw **1400** from being pulled out of the oral fluid container **1100** once inserted. In the embodiment shown in FIGS. **21A-21C**, the spittoon straw **1400** includes two projections **1496**, though in other embodiments, the spittoon straw **1400** may include more projections **1496**. Alternatively, in other embodiments, the spittoon straw **1400**, the oral fluid bottle **1101**, and/or the cylinder solution cup **1300** include a different mechanism for preventing the spittoon straw **1400** from sliding out of the oral fluid bottle **1101**. In one example, the spittoon straw **1400** includes projections **1496** further down the length of the tube **1490**, and the top of the cylinder solution cup **1300** includes notches that engage with the projections **1496**. Accordingly, once the spittoon straw **1400** is pushed into the cup **1300**, the engagement between the projections **1496** and the notches prevent the spittoon straw **1400** from being subsequently pulled out from the cup **1300**. In another example, the funnel **1416** includes projections, and the top of the oral fluid bottle **1101** includes notches that engage with the projections such that once the spittoon straw **1400** is pushed far enough into the oral fluid bottle **1101**, the notches engage with the projections and the spittoon straw **1400** cannot subsequently be pulled back beyond the notches.

(90) Furthermore, the spittoon straw **1400** includes apertures **1498** formed into a side wall of the tube **1490** of the spittoon straw **1400**. The apertures **1498** allow the oral fluid container **1100**, including the oral fluid bottle **1101**, cylinder solution cup **1300**, and spittoon straw **1400**, to be spill-resistant. For example, when the oral fluid container **1100** is tipped, liquids contained within the receptacle **1102** (e.g., unused oral fluid, brushing waste) may flow into the tube **1490** of the spittoon straw but subsequently flow out through the apertures **1498** in the side wall of the tube **1490** and back into the receptacle **1102**. The apertures **1498** are designed to provide a maximum amount of flow out of the tube **1490** without compromising the structural integrity of the tube **1490**.

(91) Referring back to FIG. **17**, the assembled oral care system **1180** is shown. Accordingly, the cylinder solution cup **1300** has been snapped into the oral fluid bottle **1101** such that the projections **1308** of the cylinder solution cup **1300** fit into the apertures **1170** of the oral fluid bottle **1101**. Additionally, the spittoon straw **1400** has been inserted into the internal chamber **1106** of the oral fluid bottle **1101** such that the projections **1496** prevent the spittoon straw **1400** from being pulled out of the oral fluid container **1100**. As shown, in the embodiment of FIG. **17**, the components of the oral fluid container **1100** (i.e., the oral fluid bottle **1101**, the cylinder solution cup **1300**, and the spittoon straw **1400**) are configured to have circular cross sections, though in other embodiments, the components may have different cross sections (e.g., elliptical, oblong).

(92) The toothbrush **1200** has also been inserted into the tube **1490** of the spittoon straw **1400** in FIG. **17**. The head **1206** of the toothbrush **1200** is positioned below the indentation **1495** such that the indentation **1495** holds the toothbrush **1200** within the spittoon straw **1400** and a small force must be applied to remove the toothbrush **1200** from the spittoon straw **1400**. Accordingly, the projection seal **1220a** rests on the base **1446** of the spittoon straw **1400**, and the projection seal **1220b** sits inside the tube **1490**. Together, the projection seals **1220a** and **1220b** seal the interior of the oral care system **1180** from outside contaminants. Additionally, the projection seal **1220a** resting on the base **1446** prevents the toothbrush **1200** from sliding completely into the tube **1490** and into the receptacle **1102**. Similarly, the projection seal **1220b** fits snugly into the tube **1490** such that the circumference of the projection seal **1220b** abuts the wall of the tube **1490**, further sealing the oral fluid container **1100**. Furthermore, the fact that the projections **1220** are offset from the center of the toothbrush **1200** allows the toothbrush **1200** to be inserted into the spittoon straw

1400 without crushing the bristles **1216** of the toothbrush head **1206**.

(93) In some embodiments, the oral care system **1180** is packaged as a pre-assembled unit as shown in FIG. 17. In other embodiments, the oral care system **1180** is packaged as separate components (e.g., the oral fluid bottle **1101**, toothbrush **1200**, cylinder solution cup **1300**, and spittoon straw **1400** are packaged separately) such that a user, such as a care provider or a patient, must assemble the oral care system **1180**. Additionally, the components of the oral care system **1180** are manufactured from any type of appropriate spill-resistant and sanitary materials. For example, in some embodiments, the seal **1314** is made from a polyethylene film, and the cylinder solution cup is a polyethylene cup. Additionally, the oral fluid bottle **1101** and the spittoon straw are made of polypropylene.

(94) Referring now to FIGS. 22A-22H, side views of the oral care system **1180** are shown during steps of use. To begin with, FIGS. 22A and 22B depict side views of the assembled oral care system **1180** before use. As shown, the point **1494** of the spittoon straw **1400** rests proximal the seal **1314** of the cylinder solution cup **1300**. To use the oral care system **1180**, a patient or care provider thus pushes on the toothbrush **1200** and/or the spittoon straw **1400** as depicted by arrow **1182** until the point **1494** of the spittoon straw **1400** breaches the seal **1314**. In some embodiments, the spittoon straw **1400** can be moved approximately 0.64 cm (0.25 inches) within the receptacle **1102**. Accordingly, the patient or care provider can move the spittoon straw **1400** in a first direction and subsequently push the spittoon straw **1400** in an opposite direction towards the seal **1314** to facilitate breaching, such as breaking, of the seal **1314**.

(95) The patient or care provider then continues to press the toothbrush **1200** and/or the spittoon straw **1400** in a direction depicted by arrow **1184** such that the opening **1492** of the spittoon straw **1400** and the head **1206** of the toothbrush **1200** become substantially submerged in the oral fluid contained within the base **1302** of the cylinder solution cup **1300**. The patient or care provider presses until the point **1494** of the spittoon straw **1400** reaches proximal the bottom of the cylinder solution cup **1300** as shown in FIGS. 22C and 22D. Doing so ensures that the head **1206** of the toothbrush **1200** is substantially submerged in the oral fluid.

(96) Once the toothbrush head **1206** is substantially submerged in the oral fluid, the patient or care provider pulls on the toothbrush **1200** and/or the spittoon straw **1400** in a direction depicted by arrow **1186**. Doing so withdraws the spittoon straw **1400** and the toothbrush **1200** contained therein. The spittoon straw **1400** continues to be withdrawn until the projections **1496** on the tube **1490** of the spittoon straw **1400** contact the outlet **1110** of the oral fluid container, as shown in FIGS. 22E and 22F. The projections **1496** thus prevent the spittoon straw **1400** from being withdrawn further. At this point, the patient or care provider continues to pull the toothbrush **1200** in a direction depicted by arrow **1188** until the head **1206** of the toothbrush **1200** slides past the indentation **1495** and out of the spittoon straw **1400**, as shown in FIG. 22G. In some embodiments, the tube **1490** is configured to squeeze excess oral fluid from the head **1206** of the toothbrush **1200** as the toothbrush **1200** is pulled out of the spittoon straw **1400**. Once the toothbrush **1200** is removed from the oral care system **1180**, the spittoon straw **1400** falls back into the oral fluid bottle **1101** and/or into the cylinder solution cup **1300** as shown in FIG. 22H. As such, a care provider can easily check that the patient has used the oral care system **1180** by looking to see whether the spittoon straw **1400** is in its original position with respect to the oral fluid container **1100**, or whether it has fallen into the oral fluid bottle as shown in FIG. 22H indicating use.

(97) Once the patient has pulled the toothbrush **1200** out of the oral care system **1180**, the patient can use the toothbrush **1200** in a standard fashion to brush the patient's teeth. During or after brushing, the patient spits into the funnel **1416** of the spittoon straw **1400** to remove oral fluid from the patient's mouth. When the patient has finished the oral care treatment, the oral care system **1180** is to be disposed. In some arrangements, the patient reinserts the toothbrush **1200** into the tube **1490** of the spittoon straw **1400** before disposing of the oral care system **1180**.

(98) Any of the oral care systems described herein, or another system combining a toothbrush (e.g.,

toothbrush **100**, toothbrush **900**, or toothbrush **1200**) with a unit dose oral fluid container (e.g., oral fluid container **200**, **400**, **500**, **600**, or **700**; oral fluid container **800** including the oral fluid bottle **801** and cylinder solution cup **1000**; or oral fluid container **1100** including the oral fluid bottle **1101**, cylinder solution cup **1300**, and spittoon straw **1400**), may be packaged individually or may be packaged as a kit including three or four oral care systems. Therefore, a care provider can leave a kit for a patient in the morning, and the patient will have the correct number of oral care systems that the patient should use in a given day. The care provider can also check the oral care systems for broken seals, brushing waste, or, in the case of the oral care system including the spittoon straw, that the spittoon straw is in the used position wherein the spittoon straw has fallen into the oral fluid container to verify patient compliance in following a preferred oral care plan. In some embodiments, the packaging may be designed with a small footprint for storage on a bedside table in a hospital room. In certain embodiments, the packaging further includes an element to cover or protect the funnel opening of the oral fluid container to minimize unwanted particles or materials from falling into the device and onto the toothbrush.

(99) The construction and arrangement of the elements of the exemplary embodiments are illustrative only. Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements. The elements and assemblies may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Additionally, in the subject description, the word “exemplary” is used to mean serving as an example, instance, or illustration. Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word “exemplary” is intended to present concepts in a concrete manner. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred and other exemplary embodiments without departing from the scope of the appended claims.

Claims

1. An oral care system comprising: a container comprising a first opening at a first end and a second opening at a second end opposite of the first end, the container defining an interior portion; a moveable funnel configured to slide through the first opening and into the interior portion of the container, wherein the moveable funnel comprises a funnel portion having a base connected to a tube, wherein a distal end of the tube includes a cutting edge angled relative to a central axis of the tube; and a toothbrush supported by the tube, the toothbrush comprising a handle extending out from the container.
2. The oral care system of claim 1, wherein movement of the toothbrush is caused by manipulation of the handle, and wherein movement of the toothbrush further causes the moveable funnel to slide in the interior portion of the container.
3. The oral care system of claim 1, further comprising a repository configured to hold a fluid and having a barrier configured to be breached.
4. The oral care system of claim 3, wherein: the second opening is configured to receive the repository; and the repository is coupled to a bottom portion of the container.
5. The oral care system of claim 3, wherein movement of the toothbrush causes the distal end of the tube to breach the barrier.

6. The oral care system of claim 3, wherein movement of the moveable funnel causes the cutting edge to breach the barrier.
 7. The oral care system of claim 1, wherein the tube includes an indentation forming a portion having a smaller diameter, thereby requiring a force to remove the toothbrush from the tube.
 8. The oral care system of claim 1, wherein the tube further comprises projections extending from an exterior surface of the tube, the projections configured to interact with an internal chamber of the container to prevent the tube from being removed from the container.
 9. The oral care system of claim 8, wherein the projections are angled such that the moveable funnel can be inserted into the internal chamber, but once inserted, the projections interact with the internal chamber of the container to prevent the tube from being removed from the container.
 10. The oral care system of claim 1, wherein the interior portion of the container includes an internal chamber, and wherein the internal chamber is configured to interact with the funnel portion of the moveable funnel to limit movement of the moveable funnel in the interior portion.
 11. The oral care system of claim 10, wherein the internal chamber comprises a second funnel portion and a neck portion.
 12. The oral care system of claim 10, wherein movement of the toothbrush by manipulation of the handle in a direction of a repository, once movement of the moveable funnel has been limited by the internal chamber, causes the toothbrush to move past the distal end of the tube into the repository.
 13. The oral care system of claim 1, wherein the toothbrush comprises a plurality of projections formed on the handle.
 14. The oral care system of claim 13, wherein at least one of the plurality of projections is configured to seal the interior portion from an exterior of the container.
 15. The oral care system of claim 1, wherein the moveable funnel further comprises apertures formed in a side wall of the tube to allow flow of fluid out of the tube and into the interior portion of the container.
 16. An oral care system comprising: a container defining an interior portion; a moveable funnel configured to slide in the interior portion of the container, wherein the moveable funnel comprises a funnel portion having a base connected to a tube, wherein a distal end of the tube includes a cutting edge angled relative to a central axis of the tube; a repository configured to hold a fluid and having a barrier configured to be breached; and a toothbrush supported by the tube, the toothbrush comprising a handle extending out from the container.
 17. The oral care system of claim 16, wherein the repository is coupled to a bottom portion of the container.
 18. The oral care system of claim 16, wherein movement of the toothbrush causes the distal end of the tube to breach the barrier.
 19. The oral care system of claim 16, wherein movement of the moveable funnel causes the cutting edge to breach the barrier.
 20. An oral care system comprising: a container defining an interior portion; a moveable funnel configured to slide in the interior portion of the container, wherein the moveable funnel comprises: a funnel portion having a base connected to a tube, wherein a distal end of the tube includes a cutting edge angled relative to a central axis of the tube, and apertures formed in a side wall of the tube to allow flow of fluid out of the tube and into the interior portion of the container; and a toothbrush supported by the tube, the toothbrush comprising a handle extending out from the container.
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